

# Setting Priors in brms

Bayesian Mixed Effects Models with brms for Linguists

Workshop Materials

2025-12-16

## Table of contents

|          |                                                                |          |
|----------|----------------------------------------------------------------|----------|
| <b>1</b> | <b>Setting Priors in brms (20 min)</b>                         | <b>1</b> |
| 1.1      | Default vs. Weakly Informative Priors . . . . .                | 1        |
| 1.1.1    | What is a Prior? . . . . .                                     | 1        |
| 1.1.2    | The Intercept Adapts to Your Data! . . . . .                   | 2        |
| 1.2      | Default brms Priors . . . . .                                  | 5        |
| 1.3      | Setting Weakly Informative Priors for Reaction Times . . . . . | 5        |
| 1.3.1    | Why These Numbers? . . . . .                                   | 6        |
| 1.3.2    | Comparison: Normal vs. Student-t . . . . .                     | 7        |
| 1.4      | Summary . . . . .                                              | 8        |

## 1 Setting Priors in brms (20 min)

### 1.1 Default vs. Weakly Informative Priors

brms uses weakly informative priors by default (not completely flat). However, for psycholinguistics, domain-specific priors are even better.

#### 1.1.1 What is a Prior?

A prior encodes your beliefs about parameter values **before** seeing the data. In Bayesian inference:

$$\text{posterior} \propto \text{likelihood} \times \text{prior}$$

**Types of priors:**

- **Flat priors:** No information - any value equally likely (bad: implies ignorance)
- **Weakly informative:** Gentle regularization - allows data to dominate
- **Domain-specific:** Based on domain knowledge - prevents unreasonable values

### 1.1.2 The Intercept Adapts to Your Data!

This is important: The default intercept prior depends on `mean(y)`. Let's see this in action:

```
library(brms)
library(tidyverse)

# Create example datasets with different scales
set.seed(123)

# RT data: log-transformed, mean 6 (400ms)
rt_data_typical <- data.frame(
  subject = factor(rep(1:20, each = 10)),
  item = factor(rep(1:10, times = 20)),
  condition = factor(rep(c("A", "B"), each = 5, times = 20)),
  log_rt = rnorm(200, mean = 6, sd = 0.5)
)

# RT data: extreme scale, mean 10 (22,000ms - unrealistic)
rt_data_extreme <- data.frame(
  subject = factor(rep(1:20, each = 10)),
  item = factor(rep(1:10, times = 20)),
  condition = factor(rep(c("A", "B"), each = 5, times = 20)),
  log_rt = rnorm(200, mean = 10, sd = 2)
)
```

Now let's check what default priors brms suggests:

```
cat("\n=== DEFAULT PRIORS: Typical RT data (mean log-RT 6) ===\n\n")
```

```
=== DEFAULT PRIORS: Typical RT data (mean log-RT 6) ===
```

```
rt_priors_typical <- get_prior(
  log_rt ~ condition + (1 + condition | subject) + (1 | item),
  data = rt_data_typical,
  family = gaussian())
```

```
)
print(rt_priors_typical)
```

|                      | prior  | class     | coef       | group   | resp | dpar | nlpar | lb | ub | tag |
|----------------------|--------|-----------|------------|---------|------|------|-------|----|----|-----|
|                      | (flat) | b         |            |         |      |      |       |    |    |     |
|                      | (flat) | b         | conditionB |         |      |      |       |    |    |     |
|                      | lkj(1) | cor       |            |         |      |      |       |    |    |     |
|                      | lkj(1) | cor       |            | subject |      |      |       |    |    |     |
| student_t(3, 6, 2.5) |        | Intercept |            |         |      |      |       |    |    |     |
| student_t(3, 0, 2.5) |        | sd        |            |         |      |      |       |    | 0  |     |
| student_t(3, 0, 2.5) |        | sd        |            | item    |      |      |       |    | 0  |     |
| student_t(3, 0, 2.5) |        | sd        | Intercept  | item    |      |      |       |    | 0  |     |
| student_t(3, 0, 2.5) |        | sd        |            | subject |      |      |       |    | 0  |     |
| student_t(3, 0, 2.5) |        | sd        | conditionB | subject |      |      |       |    | 0  |     |
| student_t(3, 0, 2.5) |        | sd        | Intercept  | subject |      |      |       |    | 0  |     |
| student_t(3, 0, 2.5) |        | sigma     |            |         |      |      |       |    | 0  |     |
| source               |        |           |            |         |      |      |       |    |    |     |
| default              |        |           |            |         |      |      |       |    |    |     |
| (vectorized)         |        |           |            |         |      |      |       |    |    |     |
| default              |        |           |            |         |      |      |       |    |    |     |
| (vectorized)         |        |           |            |         |      |      |       |    |    |     |
| default              |        |           |            |         |      |      |       |    |    |     |
| default              |        |           |            |         |      |      |       |    |    |     |
| (vectorized)         |        |           |            |         |      |      |       |    |    |     |
| (vectorized)         |        |           |            |         |      |      |       |    |    |     |
| (vectorized)         |        |           |            |         |      |      |       |    |    |     |
| (vectorized)         |        |           |            |         |      |      |       |    |    |     |
| (vectorized)         |        |           |            |         |      |      |       |    |    |     |
| default              |        |           |            |         |      |      |       |    |    |     |

```
cat("\n\n=== DEFAULT PRIORS: Extreme RT data (mean log-RT 10) ===\n\n")
```

```
=== DEFAULT PRIORS: Extreme RT data (mean log-RT 10) ===
```

```
rt_priors_extreme <- get_prior(
  log_rt ~ condition + (1 + condition | subject) + (1 | item),
  data = rt_data_extreme,
  family = gaussian())
```

```
)
print(rt_priors_extreme)
```

```

      prior      class      coef  group resp dpar nlpar lb ub tag
      (flat)         b
      (flat)         b conditionB
      lkj(1)        cor
      lkj(1)        cor          subject
student_t(3, 10, 2.5) Intercept
student_t(3, 0, 2.5)      sd                      0
student_t(3, 0, 2.5)      sd          item          0
student_t(3, 0, 2.5)      sd Intercept item          0
student_t(3, 0, 2.5)      sd          subject          0
student_t(3, 0, 2.5)      sd conditionB subject          0
student_t(3, 0, 2.5)      sd Intercept subject          0
student_t(3, 0, 2.5)      sigma                    0
      source
      default
(vectorized)
      default
(vectorized)
      default
      default
(vectorized)
(vectorized)
(vectorized)
(vectorized)
(vectorized)
      default
```

```
# Compare intercept priors
cat("\n=== Intercept Comparison ===\n")
```

```
=== Intercept Comparison ===
```

```
cat("Typical data prior: ",
    rt_priors_typical[rt_priors_typical$class == "Intercept", "prior"], "\n")
```

```
Typical data prior:  student_t(3, 6, 2.5)
```

```
cat("Extreme data prior: ",  
    rt_priors_extreme[rt_priors_extreme$class == "Intercept", "prior"], "\n")
```

Extreme data prior:    student\_t(3, 10, 2.5)

```
cat("→ The intercept prior CHANGES with data scale!\n")
```

→ The intercept prior CHANGES with data scale!

**Key insight:** The intercept prior automatically scales with your data. This is convenient but has a problem: **if you don't specify priors, your prior assumptions implicitly depend on how you code your variables!**

## 1.2 Default brms Priors

When you don't specify priors, brms assigns defaults:

- **Intercept:** `student_t(3, mean(y), 2.5)` - **DATA-DEPENDENT!** Centers at your data mean
  - Adapts to your data scale automatically
  - For RT data with `mean(log_rt) = 6`: allows roughly 150ms-1100ms range
- **Slopes (b):** `(flat)` - improper uniform prior over  $(-\infty, +\infty)$ 
  - No information: any effect size equally likely
  - Technically improper (doesn't integrate to 1)
- **Sigma** (residual SD): `student_t(3, 0, 2.5)` with lower bound 0
  - Weakly informative for residual variance
- **SD** (random effects): `student_t(3, 0, 2.5)` with lower bound 0
  - Encourages moderate between-subject/item variation
- **Cor** (correlations): `lkj(1)` - uniform over all correlation matrices

## 1.3 Setting Weakly Informative Priors for Reaction Times

For psycholinguistics, it's better to specify priors based on domain knowledge:

```
# Define priors explicitly
rt_priors <- c(
  prior(normal(6, 1.5), class = Intercept, lb = 4), # log(RT) around 400ms, min ~55ms
  prior(normal(0, 0.5), class = b),                # effects typically < 150ms
  prior(exponential(1), class = sigma),              # residual SD
  prior(exponential(1), class = sd),                 # random effects SD
  prior(lkj(2), class = cor)                         # correlations
)

cat("Our weakly informative priors:\n")
```

Our weakly informative priors:

```
print(rt_priors)
```

|  | prior          | class     | coef | group | resp | dpar | nlpar | lb   | ub   | tag | source |
|--|----------------|-----------|------|-------|------|------|-------|------|------|-----|--------|
|  | normal(6, 1.5) | Intercept |      |       |      |      |       | 4    | <NA> |     | user   |
|  | normal(0, 0.5) | b         |      |       |      |      |       | <NA> | <NA> |     | user   |
|  | exponential(1) | sigma     |      |       |      |      |       | <NA> | <NA> |     | user   |
|  | exponential(1) | sd        |      |       |      |      |       | <NA> | <NA> |     | user   |
|  | lkj(2)         | cor       |      |       |      |      |       | <NA> | <NA> |     | user   |

### 1.3.1 Why These Numbers?

#### 1.3.1.1 normal(6, 1.5) for Intercept with lb = 4

- **Mean = 6** on log scale  $\rightarrow \exp(6)$  403ms (typical RT)
- **SD = 1.5**  $\rightarrow$  95% prior interval spans [3, 9] on log scale
- **Lower bound = 4**  $\rightarrow \exp(4)$  55ms (minimum physiologically possible motor response)
  - Without lb, the prior puts only ~9% probability below this threshold
  - Adding lb makes our assumption explicit: we don't entertain impossibly fast RTs
  - In practice, data will dominate anyway, but bounded priors clarify our theoretical assumptions
- But 95% of prior **mass** is between  $\pm 1.96 \times 1.5$  around the mean
- This puts most probability on reasonable RTs (100-1500ms), while still allowing outliers

### 1.3.1.2 `normal(0, 0.5)` for Effects

- **Mean = 0** (no directional assumption)
- **SD = 0.5** on log scale
- 95% prior interval:  $[-1, 1]$  on log scale
- Translates to effect sizes of roughly  $\pm 65\%$  of baseline (multiplicative)
- Or approximately  $\pm 100\text{-}150\text{ms}$  for typical RTs around  $400\text{-}600\text{ms}$
- **Why not flat?** Flat priors prefer extreme effect sizes (counterintuitive!)

### 1.3.1.3 `exponential(1)` for Sigma and SD

- Encourages moderate variance while allowing flexibility
- Penalizes very large residual variation or random effect variance
- Mean = 1 on the scale of log-RTs

### 1.3.1.4 `lkj(2)` for Correlations

- $= 2$ : slight preference for correlations near 0
- Skeptical of strong correlations (like perfect intercept-slope correlation)
- If you truly expected strong correlations, you could use `lkj(1)` or lower

## 1.3.2 Comparison: Normal vs. Student-t

brms defaults use `student_t(3, , 2.5)` which has heavier tails than `normal()`. Why switch?

```
# Compare tail behavior
normal_99 <- quantile(rnorm(100000, 0, 0.5), c(0.001, 0.999))
studentt_99 <- quantile(rt(100000, 3) * 0.5, c(0.001, 0.999))

cat("Tails comparison (1st and 99th percentiles):\n")
```

Tails comparison (1st and 99th percentiles):

```
cat("Normal(0, 0.5):", round(normal_99, 2), "\n")
```

Normal(0, 0.5):            -1.55 1.54

```
cat("Student-t(3) × 0.5: ", round(studentt_99, 2), "\n")
```

```
Student-t(3) × 0.5:      -5.19 5.13
```

```
cat("Student-t ratio:      ", round(studentt_99[2] / normal_99[2], 2), "x wider\n\n")
```

```
Student-t ratio:        3.34 x wider
```

```
cat("Student-t allows extreme values 2x more likely than normal!\n")
```

```
Student-t allows extreme values 2x more likely than normal!
```

**When to use each:** - **Student-t:** Default choice (conservative, robust to outliers) - **Normal:** When you have strong domain knowledge about plausible ranges (better for RT data in controlled experiments)

**Good sign:** Prior predictions should cover the plausible range of your outcome variable, but not too widely.

## 1.4 Summary

### Key takeaways:

1. **Don't use flat priors** - they're uninformative and often lead to weak regularization
2. **Default intercept priors adapt to data** - implicit assumptions depend on your coding!
3. **Specify priors explicitly** based on domain knowledge
4. **Use prior predictive checks** - verify that priors generate plausible predictions before fitting
5. **Normal() is better than student\_t() when you have domain knowledge** - more concentrated around plausible values

**Next steps:** - See 02\_prior\_predictive\_checks\_rt.qmd for detailed prior validation - See 03\_posterior\_predictive\_checks\_rt.qmd for checking if the fitted model makes sense